GET 324 – Laboratory Exercise 10 (AE1 Mini Project Report)

Title: Potato Healthy vs Potato Late Blight Classification Using a Convolutional Neural Network (CNN)

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Report

This project involved the development of a Convolutional Neural Network (CNN)-based application for classifying potato leaf images into Healthy and Late Blight categories. The dataset was obtained from Kaggle and preprocessed through image resizing, normalization, and data augmentation to improve model performance and generalization. TensorFlow/Keras was used to train the CNN model, while Streamlit was used to develop an interactive web application that enables users to upload potato leaf images for instant disease detection with confidence scores. During the development and deployment stages, challenges included dataset limitations, incorrect predictions for unrelated images, model deployment, and integration of the trained model from Google Drive. These challenges were addressed by improving image preprocessing, debugging the application, using gdown for automatic model download, and refining the prediction logic. Future improvements include expanding the dataset, increasing model accuracy through advanced deep learning techniques, and incorporating additional potato diseases into the application for more comprehensive diagnosis.